

AP Precalculus: 2026–2027

Unit 4: Functions Involving Parameters, Vectors, and Matrices Lesson 4.6: Conic Sections (3 instructional periods, 55 minutes each)

Primary Objective: Students will write, interpret, and convert analytic representations of parabolas, ellipses, circles, and hyperbolas with horizontal or vertical lines of symmetry, identifying center or vertex, axes radii, and asymptotes from standard form equations. (*Big Idea: Functions*)

Priority Ideas & Skills

AP Skills

1.B Express in equivalent forms — Rewrite a general second-degree equation (e.g. $4x^2 - 9y^2 = 36$) in standard conic form by dividing through or completing the square, then identify the conic type and its key measurements.

2.A Identify information from representations — Extract the center, vertex, semi-axes, and asymptotes directly from a standard-form equation without graphing.

2.B Construct equivalent representations — Given a verbal or geometric description of a conic (center or vertex plus key measurements), write the standard-form equation.

Key Understandings (EKs)

EK 4.6.A.1: A parabola with vertex (h, k) can, if $a \neq 0$, be represented analytically as $x - h = a(y - k)^2$ if it opens left or right, or as $y - k = a(x - h)^2$ if it opens up or down.

EK 4.6.A.2: An ellipse centered at (h, k) with horizontal radius a and vertical radius b is represented as $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$. A circle is a special case where $a = b$.

EK 4.6.A.3: A hyperbola centered at (h, k) is $\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$ (opens left/right) or $\frac{(y - k)^2}{b^2} - \frac{(x - h)^2}{a^2} = 1$ (opens up/down), with asymptotes $y - k = \pm \frac{b}{a}(x - h)$.

Vocabulary, Concepts, & Theorems

Term	Definition / Note
conic section	A curve formed by the intersection of a plane and a double cone: parabola, ellipse (including circles), or hyperbola.
parabola	The set of points equidistant from a fixed focus and a fixed directrix; vertex (h, k) in standard form.
vertex	The turning point of a parabola; the point (h, k) in $y - k = a(x - h)^2$ or $x - h = a(y - k)^2$.
ellipse	The set of points for which the sum of distances to two foci is constant; has a center and two perpendicular axes.
center	The point (h, k) at the center of an ellipse or hyperbola.
semi-major / semi-minor axis	The larger (a) and smaller (b) half-axes of an ellipse.
circle	Special ellipse with $a = b = r$; equation $(x - h)^2 + (y - k)^2 = r^2$.
hyperbola	The set of points for which the <i>difference</i> of distances to two foci is constant; has two branches and two asymptote lines.
asymptote	Lines the hyperbola approaches but never reaches: $y - k = \pm \frac{b}{a}(x - h)$.

Activate Prior Knowledge & Spiral Review

Spiral topics activated during Warm-Up (first 8 minutes):

- **Completing the square:** Rewrite $x^2 - 6x$ as $(x - 3)^2 - 9$ — the essential algebraic tool for converting general equations to standard conic form.
- **Equation of a circle:** Recall $(x - h)^2 + (y - k)^2 = r^2$ as a familiar instance of the ellipse standard form, grounding new content in prior knowledge.
- **Reading standard form:** Identify center and radii from a given ellipse equation — directly previews extracting features from ellipse and hyperbola equations.

Hook — Entry Event

Scenario: “Planetary orbits are ellipses. Telescope mirrors are parabolas. Navigation systems like LORAN use hyperbolas. What do all these shapes have in common, and how do we write equations that pin them down precisely?”

Discussion prompts:

1. “A satellite dish focuses signals at one point. What shape is the dish, and why does that shape work?”
2. “Earth’s orbit is an ellipse with the Sun at one focus. How is an ellipse different from a circle?”
3. “Two radio stations broadcast simultaneously. A ship measures the difference in arrival times. The set of positions with a fixed difference in distances to the two stations is what shape?”

Key tension to surface: All three conics share elegant standard-form equations. Mastering those equations lets us describe physical curves precisely.

Transition: “Over three periods we master the standard form of each conic, identify key features, and write equations from descriptions.”

Lesson — Period 1: Parabolas

Step 1 — Two Orientations (10 min)

Present both standard forms:

- **Vertical axis:** $y - k = a(x - h)^2$ ($a > 0$ opens up; $a < 0$ opens down)
- **Horizontal axis:** $x - h = a(y - k)^2$ ($a > 0$ opens right; $a < 0$ opens left)

Emphasize: the *squared variable* tells you the axis of symmetry.

Step 2 — Reading the Equation (20 min)

Model: $y + 3 = 2(x - 1)^2$. Rewrite as $y - (-3) = 2(x - 1)^2$. Vertex $(1, -3)$; $a = 2 > 0 \Rightarrow$ opens *up*.

Student practice:

- $y - 5 = -(x + 2)^2$ — vertex and direction?
- $x - 4 = -\frac{1}{2}(y - 1)^2$ — vertex and direction?

Ask: “How does the sign of a tell you direction without graphing?”

Pacing: Reserve final 15 min for students to start Tier R portion of activity.

Lesson (cont.) — Period 2: Ellipses and Circles

Step 3 — Ellipse Standard Form (15 min)

Standard form: $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$

Example: $\frac{(x + 1)^2}{16} + \frac{(y - 3)^2}{9} = 1$

- Center: $(-1, 3)$
- $a^2 = 16 \Rightarrow a = 4$ (horizontal radius)
- $b^2 = 9 \Rightarrow b = 3$ (vertical radius)

Step 4 — Circle as Special Ellipse (15 min)

When $a = b = r$: $(x - h)^2 + (y - k)^2 = r^2$.

Confirm: $\frac{(x - 2)^2}{25} + \frac{(y + 1)^2}{25} = 1$ is a circle, center $(2, -1)$, $r = 5$.

Writing equations: Center $(3, -2)$, $a = 5$, $b = 2$. Have students write the ellipse equation, then write a circle equation from center and radius.

Pacing: Reserve 20 min for Tier A activity problems.

Lesson (cont.) — Period 3: Hyperbolas

Step 5 — Two Orientations (12 min)

- Opens left/right: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

- Opens up/down: $\frac{(y-k)^2}{b^2} - \frac{(x-h)^2}{a^2} = 1$

Key diagnostic: the *positive* squared term tells you the direction of opening.

Step 6 — Asymptotes (15 min)

From $\frac{(x-2)^2}{9} - \frac{(y+1)^2}{4} = 1$:

- Center $(2, -1)$; $a = 3$, $b = 2$

- Asymptotes: $y + 1 = \pm \frac{2}{3}(x - 2)$

Contrast with ellipse: same form but minus sign; opens rather than closes; asymptotes rather than bounded shape.

Students: $\frac{(y+3)^2}{16} - \frac{(x-1)^2}{9} = 1$ — identify center, a , b , orientation, and asymptotes.

Pacing: Exit ticket in final 8 minutes; collect for formative review.

Active Monitoring

- **Sign of h and k :** Students read $(x+1)^2$ and write center $(1, \cdot)$ instead of $(-1, \cdot)$. Prompt: “Rewrite as $(x-?)^2$ so the sign matches the formula.”
- **Horizontal vs. vertical parabola:** Students confuse which variable is squared. Prompt: “Which variable is squared? That axis is the axis of symmetry.”
- **Radii vs. a^2 :** Students report $a = 16$ instead of $a = 4$. Prompt: “The denominator is a^2 . Take the square root to find a . ”
- **Hyperbola direction:** Students pick the wrong orientation. Prompt: “Which term is positive — the x -term or the y -term? That variable’s axis is the direction of opening.”
- **Asymptote slope:** Students confuse b/a and a/b . Prompt: “Write $y - k = \pm(b/a)(x - h)$ and substitute your a and b . ”

Group Work & Differentiation

Students work in groups of 3–4 on the tiered activity (assign by teacher or allow self-selection with guidance).

Tier R — Remediate

Given equations in standard form, identify:

- conic type
- center or vertex
- key measurements (a , b , or r)

No conversion or equation-writing required.

Tier A — Approaching

Given verbal descriptions (center or vertex and key measurements), write the standard-form equation. Includes one parabola, one ellipse, and one hyperbola (with asymptotes).

Tier E — Extension

Given general-form equations (e.g. $4x^2 - 9y^2 = 36$; $x^2 + y^2 - 4x + 6y = 3$), rewrite in standard form and identify all features including asymptotes where applicable.

Individual Work & Assessment

Exit ticket administered independently (final 8 minutes of Period 3).

1. Write the standard form equation of an ellipse centered at $(3, -1)$ with horizontal radius 5 and vertical radius 2.

2. Identify the type of conic, its center, and its asymptotes (if any) for $\frac{(y+2)^2}{16} - \frac{(x-1)^2}{9} = 1$.

Data use: Students who cannot identify conic type need reinforcement reading sign patterns. Students who mix up center signs need targeted reteaching of the $(x-h)$ convention.

Reinforcement & Extension

Homework (Lesson 4.6): 6 problems covering identifying features from standard-form equations, writing equations from descriptions, both parabola orientations, and one AP-style multiple-choice item.

Extension: Given $x^2 + 4y^2 - 6x + 8y - 3 = 0$, complete the square in both variables to rewrite in standard ellipse form and identify all features. This bridges to the Tier E activity and previews algebraic techniques used in calculus.

Preview of next lesson: Ask students: “We’ve described conics with equations in x and y . Could you write parametric equations $x(t)$ and $y(t)$ that trace the same shapes? How might that be useful in modeling motion?”